Does Uncertainty Change Control? A Matched Intervention Study of LLM Action Selection in Biomedical Claim Verification

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Abstract

Code and data: https://anonymous.4open.science/r/SOEA-Plus-355B

Existing evaluations of LLM metacognition typically observe an LLM's answer, uncertainty statement, and action choice together, making it impossible to determine whether the action actually responds to uncertainty. We introduce a **matched counterfactual intervention** that holds the claim, evidence, and first-order decision fixed while varying only the uncertainty signal available at the control stage. In a pilot study on biomedical claim verification (PubMedQA), we apply this intervention—the Counterfactual Uncertainty Control Response (CUCR) framework—to two models of the same family at different scales (Llama-3.3-70b and Llama-3.1-8b), enabling a controlled comparison of scale effects on uncertainty-to-control responsiveness. The 70B model shows near-complete behavioral responsiveness in the pilot CUCR setting ($\Delta = +0.84$), while the 8B model exhibits a non-monotonic failure pattern we term the *high-uncertainty regime collapse*, where safe-action rates fall sharply at extreme uncertainty values ($\Delta = +0.18$). These findings are situated within a broader evaluation of 1,000 PubMedQA cases across three contemporary LLMs (GPT-4.1-mini, Llama-3.3-70b, Gemini-2.5-Flash), which reveals a consistent Control Collapse Gap (CCG): models commit to incorrect predictions at rates between 70.6% and 100%, despite generating uncertainty estimates. Together, these results suggest that LLM control must be tested by intervention, not inferred from co-produced uncertainty.

1 Introduction

The deployment of Large Language Models (LLMs) in safety-critical domains such as biomedicine requires robust metacognitive capabilities. A reliable model must not only generate accurate decisions but also accurately monitor its own uncertainty and, crucially, exert appropriate control—such as abstaining or seeking additional evidence—when uncertainty is high (Kadavath et al., 2022; Griot et al., 2025).

However, existing evaluations typically observe an LLM's answer, uncertainty statement, and action choice together within a single prompt or conversation. Such observations cannot determine whether the action actually responds to uncertainty, because all outputs may be co-generated by the same context. A model may report high uncertainty and still not use it to guide its actions; the missing test is counterfactual responsiveness.

In this paper, we test uncertainty-to-control responsiveness by holding the claim, evidence, and first-order decision fixed while counterfactually varying only the uncertainty information available at the control stage. This design turns uncertainty use from an inferred correlation into a directly testable behavioral outcome.

Our contributions are threefold:

1. **A Scaled Metacognitive Evaluation:** We evaluate 1,000 biomedical claim-evidence pairs from PubMedQA across three contemporary LLMs under both single-prompt (Protocol A) and multi-turn (Protocol B) elicitation, revealing a consistent Control Collapse Gap (CCG) across all models.

2. **Supporting SOEA-Plus Analysis:** We report PDEMC (a composite score integrating Decision Accuracy, Monitoring Fidelity, and Control Rationality) and weight sensitivity analysis as supporting evidence for the broader evaluation. The main contribution of this work is the CUCR intervention; PDEMC serves as a structured lens for interpreting the observational results.

3. **The CUCR Intervention Framework:** A matched counterfactual intervention that holds

claim, evidence, and decision fixed while varying only the imposed uncertainty signal. A pilot study comparing Llama-3.3-70b and Llama-3.1-8b reveals a scale-dependent divergence in control responsiveness, including a high-uncertainty regime collapse in the smaller model.

2 Related Work

2.1 Calibration and Uncertainty Estimation

The problem of model calibration has received substantial attention. Guo et al. (2017) demonstrated that modern neural networks are systematically overconfident, proposing temperature scaling as a post-hoc remedy. Jiang et al. (2021) showed that verbalized confidence in language models does not reliably correlate with accuracy. Tian et al. (2023) further demonstrated that instruction-tuned models exhibit complex calibration patterns that vary by task type, while Xiong et al. (2024) evaluated black-box confidence elicitation, finding that models tend toward overconfidence but improve with scale. These works evaluate whether uncertainty estimates are calibrated, but they do not determine whether the model's downstream action changes when uncertainty alone is manipulated.

2.2 Selective Prediction and Abstention

Selective prediction (Geifman and El-Yaniv, 2017) allows models to abstain when uncertain, trading coverage for accuracy. Kamath et al. (2020) applied this to question answering under domain shift, demonstrating that softmax probabilities alone are insufficient. Varshney et al. (2022) evaluated selective prediction across numerous datasets, finding that methods do not consistently outperform MaxProb. Recently, Presacan et al. (2026) reviewed LLM abstention in healthcare, proposing a decision-theoretic framework for safety-driven abstention. Selective prediction studies answer whether a model should answer or abstain; our setting generalizes control to multiple actions and tests uncertainty responsiveness through a matched intervention.

2.3 Metacognition and the Knowing-Doing Gap

Kadavath et al. (2022) introduced self-evaluation as a metacognitive probe, finding that large models can assess their own correctness. However, Griot et al. (2025) demonstrated that current LLMs exhibit a critical disconnect between perceived and actual capabilities in medical reasoning. Wang (2026) identified a “knowing-doing gap” in LLMs, showing that providing models their own uncertainty does not reliably improve their control decisions. Zhang et al. (2026) argued theoretically that calibration is not control, calling for intervention-based evaluations. Closest to our work, these two studies argue that self-knowledge and calibration do not guarantee appropriate action. CUCR differs by introducing a matched intervention in biomedical claim verification, holding the claim, evidence, and first-order decision fixed while varying only the uncertainty signal.

2.4 Biomedical Claim Verification

We evaluate our framework on PubMedQA (Jin et al., 2019), a dataset of expert-labeled biomedical research questions. Similar tasks include SciFact (Wadden et al., 2020), which verifies scientific claims against research abstracts. Biomedical claim verification provides a high-stakes testbed where incorrect commitment under uncertainty is especially consequential. To our knowledge, no prior work has directly tested uncertainty-to-control responsiveness in biomedical claim verification using a matched intervention that holds claim, evidence, and first-order decision fixed while varying only the uncertainty signal.

3 Methodology

3.1 The SOEA-Plus Three-Stage Protocol

SOEA-Plus evaluates LLMs through three sequential stages applied to each biomedical claim-evidence pair:

Stage 1 — Decision. The model classifies the claim as SUPPORTED, REFUTED, or INCONCLUSIVE based on the provided evidence.

Stage 2 — Monitoring. The model provides an error probability $p \in [0, 1]$, where $p = 0$ indicates certainty of correctness and $p = 1$ indicates certainty of error.

Stage 3 — Control. Based on its decision and error probability, the model selects one of four explicitly defined actions:

- **COMMIT:** Proceed with the current decision; appropriate when uncertainty is low and the decision is reliable.
- **REVISE:** Change the current decision; appropriate when evidence appears to contradict the

decision.

- **ABSTAIN:** Decline to provide a decision; appropriate when uncertainty is too high to commit safely.
- **SEEK_EVIDENCE:** Request additional evidence; appropriate when current evidence is insufficient or ambiguous.

We evaluate this under two protocols: **Protocol A** (single-prompt, all stages elicited jointly, N=1,000) and **Protocol B** (multi-turn, stages elicited sequentially to prevent self-conditioning bias, N=200).

3.2 The PDEMC Metric

The PDEMC score integrates all three stages:

$$\text{PDEMC} = 0.30 \times \text{Acc} + 0.30 \times \text{MF} + 0.40 \times \text{CR} \quad (1)$$

where Acc is Stage 1 accuracy, MF (Monitoring Fidelity) is $1 - \text{MAE}(p_i, e_i)$, and CR (Control Rationality) is defined as:

$$\text{CR} = \begin{cases} 1 & \text{if } (\hat{y}_i = y_i \wedge a_i = \text{COMMIT}) \\ & \vee (\hat{y}_i \neq y_i \wedge a_i \in \{\text{REVISE}, \text{ABSTAIN}, \text{SEEK}\}) \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Control receives the highest weight (40%) as the safety-critical output, while Accuracy and Monitoring Fidelity are weighted equally (30% each) to balance first-order correctness with self-awareness. Sensitivity analysis over four alternative weight configurations confirms that model rankings are stable across all configurations (see Section 4.3).

3.3 The Control Collapse Gap

The Control Collapse Gap (CCG) quantifies the rate at which models commit to incorrect predictions:

$$\text{CCG} = \frac{|\{i : \hat{y}_i \neq y_i \wedge a_i = \text{COMMIT}\}|}{|\{i : \hat{y}_i \neq y_i\}|} \quad (3)$$

A high CCG indicates that even when a model is wrong, it overwhelmingly chooses to commit rather than abstain or revise.

3.4 The CUCR Intervention

To provide intervention-based behavioral evidence that the CCG reflects a failure of control rather than merely poor calibration, we introduce the Counterfactual Uncertainty Control Response (CUCR) framework. For each evaluated sample, we hold the claim, evidence, and first-order decision fixed, and inject an imposed uncertainty level $U_{\text{imposed}} \in \{0.10, 0.30, 0.50, 0.70, 0.90\}$ into the prompt, asking the model to select a control action given this specific value. These five levels were chosen to provide evenly spaced intervals covering the full probability range. We also include a *masked* condition in which no uncertainty value is provided, serving as a baseline. If the model behaviorally uses the supplied uncertainty signal, its Safe Action Rate should rise monotonically with U_{imposed}.

The pilot CUCR study (N=50 per model, chosen to establish initial behavioral bounds before larger-scale deployment) specifically targets a *scale-associated comparison*: we contrast Llama-3.3-70b (70B parameters) against Llama-3.1-8b (8B parameters) from the same model family, primarily varying scale. We measure responsiveness via $\Delta = (U = 0.9) - (U = 0.1)$, capturing the maximum behavioral shift across the uncertainty spectrum. We note that differences between the two models may reflect factors beyond parameter count, including instruction-tuning and serving configurations.

3.5 Dataset and Models

We evaluate on 1,000 claim-evidence pairs sampled from PubMedQA (Jin et al., 2019) (seed=42, `pqa_labeled` split), with labels mapped from {yes, no, maybe} to {SUPPORTED, REFUTED, INCONCLUSIVE}. The same 1,000 cases are used across all models and conditions. Main SOEA-Plus evaluation uses GPT-4.1-mini, Llama-3.3-70b, and Gemini-2.5-Flash; the CUCR pilot uses Llama-3.3-70b and Llama-3.1-8b.

4 Main Results

4.1 Protocol A: Control Collapse Across Models (N=1,000)

Table 1 presents the main results. GPT-4.1-mini achieves the highest accuracy (46.2%) but exhibits a CCG of 70.6%. Llama-3.3-70b shows a CCG of 91.2%, while Gemini-2.5-Flash commits to every incorrect prediction (CCG = 100%), completely ignoring its own mean error probability of 0.500. Figure 1 visualizes the PDEMC component breakdown, highlighting that low Control Rationality (CR) is the primary driver of poor overall performance across all models.

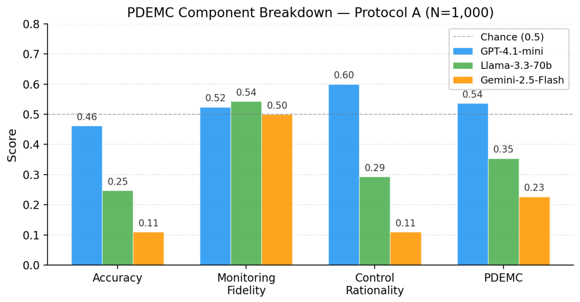

Figure 1: PDEMC component breakdown (Protocol A). While Accuracy and Monitoring Fidelity vary, Control Rationality remains the weakest component for all models except GPT-4.1-mini.

Model	Acc	Safe	CCG	PDEMC
GPT-4.1-mini	0.462	0.178	70.6%	0.536
Llama-3.3-70b	0.247	0.086	91.2%	0.354
Gemini-2.5-Flash	0.110	0.000	100.0%	0.227

Table 1: Main results, Protocol A (N=1,000, seed=42). Safe = safe action rate (ABSTAIN + REVISE + SEEK_EVIDENCE). 95% CIs: GPT Acc [0.432–0.491], Llama Acc [0.222–0.276], Gemini Acc [0.092–0.131].

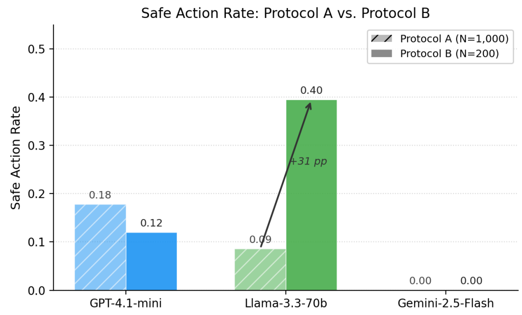

Figure 2: Safe Action Rate comparison between Protocol A (single-prompt) and Protocol B (multi-turn). Llama-3.3-70b exhibits a dramatic +31 percentage point improvement under staged elicitation.

Model	Protocol	Acc	Safe	Commit
GPT-4.1-mini	A (N=1,000)	0.462	0.178	0.822
	B (N=200)	0.460	0.120	0.880
Llama-3.3-70b	A (N=1,000)	0.247	0.086	0.914
	B (N=200)	0.425	0.395	0.605
Gemini-2.5-Flash	A (N=1,000)	0.110	0.000	1.000
	B (N=200)	0.075	0.000	1.000

Table 2: Protocol A (N=1,000) vs. Protocol B (N=200). Protocol B was applied to a 200-case subset due to its multi-turn cost. Acc = accuracy, Safe = safe action rate (ABSTAIN + REVISE + SEEK), Commit = overall commit rate (= 1 − Safe). Note: Commit rate differs from CCG (Table 1), which conditions on incorrect predictions only.

4.2 Protocol Comparison (N=200)

Table 2 compares single-prompt (Protocol A, N=1,000) vs. multi-turn (Protocol B, N=200) elicitation. Because Protocol B requires three sequential API calls per case, it was applied to the first 200 cases of the dataset; Protocol A numbers in this table reflect the full N=1,000 evaluation. As visualized in Figure 2, Llama-3.3-70b shows a substantial improvement under Protocol B: accuracy rises from 0.247 to 0.425, and the safe action rate increases from 8.6% to 39.5% (+31 percentage points). This suggests that staged elicitation, which forces the model to articulate its uncertainty before selecting an action, meaningfully improves control responsiveness. GPT-4.1-mini and Gemini-2.5-Flash show minimal protocol sensitivity.

4.3 Sensitivity Analysis

Table 3 reports PDEMC scores under four alternative weight configurations. Model rankings remain identical across all configurations, confirming that the PDEMC metric is robust to reasonable weight perturbations.

4.4 Error Analysis (N=3,600)

A comprehensive audit of 3,600 evaluated rows (3 models × 1,000 Protocol A + 3 models × 200 Protocol B) categorized five failure types (Table 4). The dominant failure mode is “Uncertainty ignored” (48.6%)—defined as committing to an incorrect prediction despite reporting an error probability $p \geq 0.30$—confirming that models frequently fail to translate their uncertainty estimates into appropriate control actions. “High-confidence wrong commit” (15.3%)—where the model makes an incorrect prediction, reports low uncertainty ($p < 0.30$), and commits—represents the most critical safety risk in high-stakes settings.

5 The CUCR Pilot Intervention

5.1 Results: Natural, Masked, and Intervention Conditions

Table 5 presents the complete CUCR results across all conditions. The *natural* condition shows the baseline safe action rate from joint elicitation; the *masked* condition shows the rate when no uncertainty value is provided; the intervention conditions show the rate at each imposed uncertainty level.

Weights (Acc/MF/CR)	GPT	Llama	Gemini
30/30/40 (original)	0.536	0.354	0.227
33/33/34 (equal)	0.529	0.360	0.239
25/25/50 (control-heavy)	0.547	0.344	0.208
50/25/25 (accuracy-heavy)	0.512	0.333	0.208

Table 3: PDEMC sensitivity analysis. Rankings are stable across all configurations.

Failure Type	Count	%
Uncertainty ignored	1749	48.6%
Safe correct commit	935	26.0%
High-confidence wrong commit	549	15.3%
Safe response to wrong decision	291	8.1%
Over-cautious safe action	76	2.1%

Table 4: Error analysis of 3,600 evaluated rows.

5.2 Scale-Dependent Divergence in Control Responsiveness

The 70B model exhibits near-complete behavioral responsiveness: at $U = 0.10$ it commits (Safe Rate = 16%), and at $U \geq 0.30$ it consistently selects safe actions (Safe Rate = 100%), maintaining this behavior across all higher uncertainty levels ($\Delta = +0.840$, 95% CI [0.74, 0.92]). The masked condition (Safe Rate = 26%) confirms that this shift is driven by the uncertainty signal rather than the prompt structure alone.

The 8B model, by contrast, exhibits a non-monotonic pattern. While it responds appropriately at moderate uncertainty ($U = 0.30$–0.70), its safe action rate collapses to 34% at extreme uncertainty ($U = 0.90$), yielding a substantially weaker $\Delta = +0.180$ (95% CI [0.02, 0.34]). We refer to this as the **high-uncertainty regime**: a region where the model appears to lose its ability to follow the uncertainty signal, potentially reverting to a pre-trained bias toward confident commitment. This finding suggests that model scale is a meaningful factor in uncertainty-to-control responsiveness, beyond its effect on first-order accuracy.

6 Discussion

Our results indicate that explicit uncertainty and behavioral control are separable. The CUCR intervention provides direct behavioral intervention evidence that while the 70B model can align its actions with an imposed uncertainty signal, the 8B model does so only within a moderate uncertainty range. This finding suggests, under the evaluated conditions, that control responsiveness is scale-dependent and non-linear.

Model	Nat.	Mask.	0.1	0.3	0.5	0.7	0.9	Δ
Llama-3.3-70b	.22	.26	.16	1.0	1.0	1.0	1.0	+.84
Llama-3.1-8b	.16	.18	.16	1.0	1.0	.96	.34	+.18

Table 5: CUCR pilot results (N=50 per model). Safe Action Rate across Natural, Masked, and five imposed uncertainty levels. Δ = (U=0.9) − (U=0.1). 95% CIs: Llama-70b Δ [0.74, 0.92]; Llama-8b Δ [0.02, 0.34].

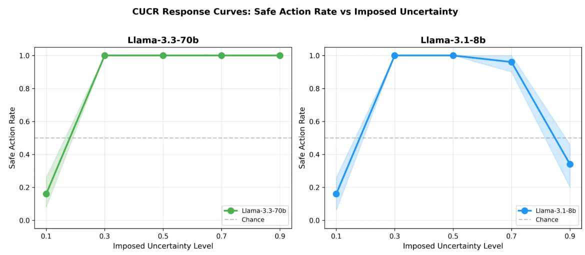

Figure 3: CUCR response curves. Llama-3.3-70b shows near-complete behavioral responsiveness. Llama-3.1-8b exhibits a high-uncertainty regime collapse at $U = 0.90$.

The stark contrast in Llama-3.3-70b's behavior between Protocol A and Protocol B further suggests that control responsiveness is highly protocol-moderated. This underscores the need for intervention-based evaluations rather than relying solely on co-produced uncertainty statements. Gemini-2.5-Flash's degenerate behavior (CCG = 100% in natural settings) further highlights that some models appear to completely decouple their stated uncertainty from their action policy.

7 Limitations

This study evaluates behavioral responsiveness, not mechanistic internal cognition. The primary testbed is biomedical claim verification (PubMedQA), and results may vary in other domains. The evaluation relies on API-based models, which are subject to versioning changes. The CUCR pilot uses N=50 samples per model; future work should scale this to the full 1,000-sample set and extend the intervention to additional model families. The intervention manipulates only the uncertainty signal while leaving the semantic content of the claim and evidence unchanged; future work should explore whether manipulating other contextual signals produces similar effects. Human-level PDEMC baselines are not directly measured and are left for future annotation studies.

Data Availability

All code, data, and results are available at `https://anonymous.4open.science/r/SOEA-Plus-355B`.

8 Conclusion

LLM control must be tested by intervention, not inferred from co-produced uncertainty. Our matched counterfactual intervention reveals a significant gap between uncertainty awareness and control responsiveness, with a scale-dependent divergence: the 70B model shows near-complete behavioral responsiveness in the pilot setting, while the 8B model exhibits a high-uncertainty regime collapse. The prevalence of "Uncertainty ignored" failures (48.6%) across the broader 1,000-sample evaluation demonstrates that calibration alone is insufficient as an evaluation criterion for high-stakes deployment.

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